

Banula Kumarage

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Summary

Innovative engineer and researcher with experience in distributed systems, cloud-native technologies, and scalable backend development. Specializing in Java(Spring Boot), Kubernetes, and real-time data processing technologies such as Apache Beam and Kafka. Skilled professional bridging academic research and industry expertise to deliver effective solutions.

Working Experience

Senior Software Engineer, Sysco LABS Sri Lanka

Feb 2026 – Present

- Led the modernization initiative to streamline the Product Catalog Building Process using Kafka Streams and Apache Flink pipelines.
- Designing a real-time data pipeline to update the Elasticsearch cluster, enabling search capabilities for APIs.
- Led the initiative to expose French translation attributes, contributing throughout the project lifecycle from planning and design to production deployment.
- Collaborated with external stakeholders to plan, coordinate, and deliver multiple cross-functional initiatives.

Software Engineer, Sysco LABS Sri Lanka

June 2024 – Feb 2026

- Designed and developed near real-time ETL pipelines, enabling data changes to be reflected within seconds.
- Led the Kinesis-to-Kafka migration initiative by planning and implementing necessary changes, and successfully executing the production cutover without downtime.
- Executed over 20 production and staging deployments with zero rollback incidents.
- Enhanced application reliability and code quality by integrating TestContainers into CI/CD pipelines for isolated integration testing.
- Conducted in-depth analysis to identify cost-saving opportunities and performance bottlenecks.

Software Engineer Intern, IFS R&D International (Pvt.) Ltd

Jan 2023 – June 2023

- Automated the manual translation rejection sync process by developing a Jenkins pipeline, reducing execution time from nearly a day to just 15 minutes.
- Investigated custom autoscaling for IFS services on Kubernetes by identifying relevant dynamic metrics using Prometheus, Grafana, Postman, and K6 load testing.
- Optimized database performance 30% through implementing a priority-based caching algorithm in the PL/SQL layer to reduce overhead.
- Implemented graceful shutdown in Java microservices for Kubernetes pods using lifecycle hooks and Dockerfile modifications, improving autoscaling efficiency and preventing request loss.

Education

Master of Science (Major Component of Research), Reading

2025 - Present

Department of Computer Science and Engineering, University of Moratuwa

- Focusing the research on mitigating the cold start problem in Serverless Platforms.
- Working on a solution that balances performance and resource utilization.
- Implementing and benchmarking the proposed solution using Apache OpenWhisk.

BSc. Eng. (Hons) Specialized in Computer Science & Engineering

2020 - 2024

Department of Computer Science and Engineering, University of Moratuwa

- Overall GPA: First Class GPA of 3.96 out of 4.2
- Dean's List in semesters 1, 2, 3, 4, 5, 6, 7, 8

- Island Rank - 173rd
- District Rank - 4th
- Z-Score - 2.2588

Publications

Use of Federated Learning for Personal Smart Media Cloud Solutions

Aug 2004

Banula Kumara, Hiruna Harankahadeniya, Akila Induranga, Indika Perera, Kutila Gunasekera
10.1109/MERCon63886.2024.10688883

Projects

Product Information Service Modernization

- Modernized the product information update and product catalog building pipeline by replacing periodic batch processing with an event-driven architecture powered by Kafka Streams, enabling near real-time data synchronization. Significantly improved application throughput and scalability by leveraging Apache Beam's parallel and distributed processing capabilities with a multithreaded execution model.
- **Technologies:** Apache Beam, Kafka Streams, Spring Boot

PicsSmart : Privacy Ensured Smart Media Cloud with Federated Learning

- An edge-processing-based application where users are concerned about their photos' privacy can still benefit from ML-powered smart features such as face-identification, face-clustering, and image caption generation. For the models to work well with heterogeneous datasets, models are to be frequently trained at the edge and updated using Federated Transfer Learning techniques.
- **Technologies:** FastAPI, ReactJS, LAVIS, Keras, Tensorflow Federated, Docker

Scatterrr : Secure and trusted distributed file storage system

- A secure distributed file storage system that segments files into equal-sized chunks and distributes them randomly across a cluster of nodes for enhanced security. Using a Merkle Tree, it ensures data integrity, allowing users to verify files remain tamper-free. The system offers an intuitive interface for uploading, searching, viewing metadata, verifying integrity, and downloading files, combining reliability with robust security principles.
- **Technologies:** Java(Spring Boot), TypeScript(Next.js)

Streamix : Dash compliant video streaming system

- The project is a DASH (Dynamic Adaptive Streaming over HTTP) compliant video streaming system. An Android app is used to capture the video and upload it to the centralized server. Anyone connected to the server can watch the uploaded videos in different qualities.
- **Technologies:** Java, Node.JS, MongoDB, FFmpeg

Moodle plugin for establishing workflows

- The purpose of the project is to create a plugin to automate the process of submitting and validating student requests regarding various Moodle activities.
- **Technologies:** PHP, MySQL, JS, Mustache

Student Request Management System

- A web-based management system to manage student requests which have been sent to the academic staff in the university.
- **Technologies:** Node.JS, Express, React, MongoDB

Skills

Technical Skills

- Java | Python | Scala | Javascript | Typescript | PHP
- Spring Boot | Django | Quarkus | FastAPI | Node.js | Express | React JS
- MySQL | MongoDB | PostgreSQL | PL/SQL | SQLite

- Kubernetes | Helm | Docker | Kafka | Apache Beam | Redis | Elasticsearch
- AWS | Azure | GCP

Non Technical Skills

- Leadership | Teamwork | Communication | Public Speaking | Analytical Thinking

Certifications

Ultimate AWS Certified Solutions Architect Associate	2026
• Completed a course on the Udemy platform	
Introduction to GCP Infrastructure	2024
• Obtained a verified badge after completing the lab tasks offered by Google Cloud	
Application Development using Microservices and Serverless	2024
• Obtained a verified badge after completing the course offered by Coursera	
AZ-900: Microsoft Azure Fundamentals Certification	2023
• Obtained a verified certificate after completing the certification exam offered by Microsoft	
Kubernetes Certified Application Developer (CKAD) with Tests	2023
• Completed a course on the Udemy platform offered by KodeCloud	

Achievements

Virtual Colleagues' Choice Award for the FY25 Sysco Technology Hackathon	Oct 2024
• Our team ByteChef has won the Virtual Colleagues Choice Award in the Sysco Global Hackathon 2024. Also, we've been selected as a Top 5 finalist for our Generative AI-based solution for boosting the productivity of our sales reps.	
Winners - RealHack 5.0	April 2024
• In the 24-hour hackathon organized by the University of Kelaniya, our implemented system of Multi-modal authenticity validation system for virtual interviews secured first place.	
IEEEExtreme 16.0 - 24-hour Algorithmic Coding Competition	Oct 2022
• Team404 World Rank – 300 Country Rank – 12	

Responsibilities

Teaching Assistant
• Computer Networks (CS3033), Data Communication (CS2033) - Department of Computer Science & Engineering, University of Moratuwa
Deputy Head - Industrial Relations Committee and Department Coordinator
• EXMO 2023 Technical Exhibition organized by the University of Moratuwa
Field Representative
• Department of Computer Science and Engineering - batch 19 - Semester 3&4

Organizations

Toastmasters International	03/2019 - Present
• Member no: 7317863	

References

- Available on Request